

TECHNOLOGY STUDENT ASSOCIATION WORK LOG - STATES TO NATIONALS CONTINUED

Supplement 1

DATE	TASK	TIME	INVOLVED TEAM MEMBER RESPONSIBLE	COMMENTS
05/02/26	Post-states redesign planning session	2 hr	OH, HW, SC, AM, AR	Reviewed judge feedback and divided Nationals redesign priorities across winch, probe, electronics, website, display, and portfolio work.
05/03/26	Website redesign planning	3 hr	AM	Mapped the Nationals website structure, mission playback needs, calibration screens, and project evidence pages.
05/04/26	In-person build and inspection session	2 hr	OH, HW, SC, AM, AR	Checked the boat, probe, wiring, and display materials together before starting major redesign work.
05/05/26	Winch issue teardown	2 hr	SC	Inspected winch wear points, spool alignment, motor connection, and tether rubbing after state testing.
05/05/26	Trifold content map	3 hr	OH	Outlined the Nationals trifold sections, image placement, build story, and technical evidence flow.
05/06/26	Probe redesign constraints review	2 hr	AR	Identified serviceability issues in the sealed probe body and listed changes needed for removable sensors.
05/06/26	Dashboard interface updates	3 hr	AM	Reworked website controls, status areas, and mission information displays for clearer operator use.
05/07/26	Shaft coupling and cable-gland research	2 hr	SC, AR	Researched shaft couplings, flexible adapters, cable glands, compression seals, and waterproof pass-through options for redesigns.
05/08/26	Full-system testing session	2 hr	OH, HW, SC, AM, AR	Ran a hands-on test session to observe winch behavior, probe handling, sensor connections, and operator workflow.
05/09/26	Cable gland research for probe	2 hr	AR	Compared gland sizes, thread options, compression ranges, and sensor-fit requirements for the redesigned probe openings.
05/09/26	PCB schematic cleanup	4 hr	AM	Updated the PCB schematic to better organize power, communication, sensor headers, and labeling.
05/10/26	Winch spool redesign	2 hr	SC	Adjusted spool geometry and cable path to reduce rubbing and improve tether payout reliability.
05/10/26	Trifold visual layout	3 hr	OH	Built the trifold layout around problem, solution, prototype, testing, and results sections.
05/11/26	Probe CAD cable-gland openings	2 hr	AR	Modified the probe body openings so cable glands could secure sensors without permanent silicone bonding.
05/12/26	Winch mount redesign	2 hr	SC	Raised and reinforced the winch mounting position to create more clearance for the wrapped tether.
05/13/26	PCB routing revision	4 hr	AM	Routed the board traces, checked component spacing, and refined silkscreen labels for assembly.
05/14/26	Winch print preparation	2 hr	SC	Prepared updated spool, mount, and coupling parts for printing and fit testing.
05/15/26	Probe serviceability revision	2 hr	AR	Refined the removable probe sections, tether entry point, and sensor access for easier maintenance.
05/16/26	Trifold graphics and photo selection	3 hr	OH	Selected build photos, diagrams, and testing visuals for the display board and cleaned up captions.
05/17/26	Sensor calibration routine planning	3 hr	AM	Planned the calibration workflow for dissolved oxygen, pH, turbidity, and supporting water-quality readings.
05/18/26	Mechanical support build session	2 hr	HW	Helped check hull balance, winch clearance, tool setup, and fit of redesigned mechanical parts.
05/21/26	Winch redesign test and revision	2 hr	SC	Tested the updated winch, checked tether tracking on the spool, and made small clearance changes.
05/22/26	Calibration testing with probe	2 hr	AM, AR	Worked through sensor calibration with the probe electronics and checked readings for consistency.
05/23/26	Portfolio state-feedback updates	3 hr	OH	Added post-states improvements, redesigned subsystem descriptions, and stronger explanation of design decisions.
05/24/26	Data parser and export refinement	3 hr	AM	Improved website data handling so mission records could be converted into clearer graphs and tables.
05/28/26	Mission playback website updates	3 hr	AM	Built playback and visualization improvements for depth, location, sensor readings, and mission review.

DATE	TASK	TIME	INVOLVED TEAM MEMBER RESPONSIBLE	COMMENTS
05/30/26	Portfolio results updates	3 hr	OH	Reworked results pages to connect testing evidence to the final engineering requirements.
06/06/26	Portfolio figures and captions	3 hr	OH	Updated figure placement, captions, photo labels, and section transitions throughout the portfolio.
06/12/26	Video asset collection	3 hr	AM	Gathered prototype clips, test footage, website captures, and subsystem images for the final video.
06/14/26	Trifold print preparation	2 hr	OH	Finalized trifold text, image alignment, heading consistency, and print-ready formatting.
06/18/26	Final in-person integration test	2 hr	OH, HW, SC, AM, AR	Tested the Nationals-version system together and checked winch, probe, website, display, and presentation readiness.
06/19/26	Final portfolio proofreading	2 hr	OH	Proofread final portfolio pages, work-log wording, image captions, and formatting before submission.
06/20/26	Final video edit and export	4 hr	AM	Edited the final video, aligned narration and visuals, and exported the finished presentation file.